

$$\begin{aligned}
 & \cdot 2+(0.2)=2+2=2 & \cdot 2+(1.0)=2+1=1 \\
 & (2+0) \cdot (2+2)=0.2=2 & (2+1) \cdot (2+0)=1.0=1 \\
 & \cdot 2+(1.1)=2+1=1 & \cdot 2+(1.2)=2+2=2 \\
 & (2+1) \cdot (2+1)=1.1=1 & \cdot (2+1) \cdot (2+2)=1.2=2 \\
 & \cdot 2+(2.0)=2+2=2 & \cdot 2+(2.1)=2+2=2 \\
 & (2+2) \cdot (2+0)=2.0=2 & \cdot (2+2) \cdot (2+1)=2.1=2 \\
 & & \cdot 2+(2.2)=2+2=2 \\
 & & (2+2) \cdot (2+2)=2.2=2
 \end{aligned}$$

$\therefore$ The distributive law $x+(y \cdot z) = (x+y) \cdot (x+z)$ holds.

~~$$\begin{aligned}
 & 0 \cdot (0+0) = 0 \cdot 0 = 0 & 0 \cdot (0+1) = 0 \cdot 0 = 0 \\
 & 0 \cdot (0+2) = 0 \cdot 0 = 0 & 0 \cdot (0+0.1) = 0+1 = 0 \\
 & 0 \cdot (0+1) = 0 \cdot 0 = 0 & 0 \cdot (1+0) = 0 \cdot 0 = 0 \\
 & 0 \cdot (0+2) = 0 \cdot 0 = 0 & 0 \cdot 1+0 \cdot 0 = 1+0 = 0 \\
 & 0 \cdot (1+1) = 0 \cdot 1 = 1 & 0 \cdot (1+2) = 0 \cdot 1 = 1 \\
 & 0 \cdot 1+0 \cdot 1 = 1+1 = 1 & 0 \cdot 0+0 \cdot 2 = 1+2 = 1 \\
 & 0 \cdot (2+0) = 0 \cdot 0 = 0 & 0 \cdot (2+1) = 0 \cdot 1 = 1 \\
 & 0 \cdot 2+0 \cdot 0 = 2+0 = 0 & 0 \cdot 0+0 \cdot 1 = 2+1 = 1 \\
 & 0 \cdot (2+2) = 0 \cdot 2 = 2 & 1 \cdot (0+0) = 1 \cdot 0 = 1 \\
 & 0 \cdot 2+0 \cdot 2 = 2+2 = 2 & 1 \cdot 0+1 \cdot 0 = 1+1 = 1 \\
 & 1 \cdot (0+1) = 1 \cdot 0 = 1 & 1 \cdot (1+0) = 1 \cdot 0 = 1 \\
 & 1 \cdot 0+1 \cdot 1 = 1+1 = 1 & 1 \cdot 1+1 \cdot 0 = 1+1 = 1 \\
 & 1 \cdot (0+2) = 1 \cdot 0 = 1 & 1 \cdot (1+1) = 1 \cdot 1 = 1 \\
 & 1 \cdot 0+1 \cdot 2 = 1+2 = 1 & 1 \cdot 1+1 \cdot 1 = 1+1 = 1 \\
 & 1 \cdot (1+2) = 1 \cdot 1 = 1 & 1 \cdot (2+0) = 1 \cdot 0 = 1 \\
 & 1 \cdot 1+1 \cdot 2 = 1+2 = 1 & 1 \cdot 2+1 \cdot 0 = 2+1 = 1 \\
 & 1 \cdot (2+1) = 1 \cdot 1 = 1 & 1 \cdot (2+2) = 1 \cdot 2 = 2 \\
 & 1 \cdot 2+1 \cdot 1 = 2+1 = 1 & 1 \cdot 2+1 \cdot 2 = 2+2 = 2 \\
 & 2 \cdot (0+0) = 2 \cdot 0 = 2 & 2 \cdot (0+1) = 2 \cdot 0 = 2 \\
 & 2 \cdot 0+2 \cdot 0 = 2+2 = 2 & 2 \cdot 0+2 \cdot 1 = 2+2 = 2 \\
 & 2 \cdot (0+2) = 2 \cdot 0 = 2 & 2 \cdot (1+0) = 2 \cdot 0 = 2 \\
 & 2 \cdot 0+2 \cdot 2 = 2+2 = 2 & 2 \cdot 1+2 \cdot 0 = 2+2 = 2
 \end{aligned}$$~~